Exercises: Spectral Theory, SVD, and Matrix Decompositions – Answer Key

Mathematical Foundations for Machine Learning

Eigenvalues, Eigenvectors, and Diagonalization

Exercise 1.1 [Easy]

Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$.

a) Find all eigenvalues and eigenvectors of A.

b) Is A diagonalizable? If so, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

c) Using the diagonalization, compute A^{10} (leave your answer in terms of P, D^{10}, P^{-1}, but compute D^{10} explicitly).

Solution

a) Characteristic polynomial: $\det(A-\lambda I) = (4-\lambda)(3-\lambda)-2 = \lambda^2-7\lambda+10 = (\lambda-5)(\lambda-2)$. So $\lambda_1 = 5$, $\lambda_2 = 2$.

For $\lambda_1 = 5$: $(A-5I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} v = 0$, so $v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

For $\lambda_2 = 2$: $(A-2I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} v = 0$, so $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

b) Yes, A has two distinct eigenvalues in $\mathbb{R}^2$, so it is diagonalizable.

$$P = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$

c) $A^{10} = PD^{10}P^{-1}$ where $D^{10} = \begin{pmatrix} 5^{10} & 0 \\ 0 & 2^{10} \end{pmatrix} = \begin{pmatrix} 9765625 & 0 \\ 0 & 1024 \end{pmatrix}$.

Exercise 1.2 [Easy]

True or False (with proof or counterexample):

a) If $A \in \mathbb{R}^{n\times n}$ has n distinct eigenvalues, then A is diagonalizable.

b) If A is diagonalizable, then A must have n distinct eigenvalues.

c) If $A^2 = A$, then the only possible eigenvalues of A are 0 and 1.

Solution

a) True. If A has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$, then the corresponding eigenvectors $v_1, \ldots, v_n$ are linearly independent (standard theorem). Since n linearly independent eigenvectors in $\mathbb{R}^n$ form a basis, A is diagonalizable.

b) False. Counterexample: $A = I_n$ (the identity matrix). It is diagonalizable (it is already diagonal), but has only one eigenvalue $\lambda = 1$ with multiplicity n.

c) True. If λ is an eigenvalue with eigenvector $v \neq 0$, then $Av = \lambda v$, so $A^2v = \lambda^2 v$. But $A^2 = A$, so $Av = \lambda^2 v$, giving $\lambda v = \lambda^2 v$, hence $(\lambda^2 - \lambda)v = 0$. Since $v \neq 0$, we get $\lambda^2 - \lambda = 0$, i.e., $\lambda(\lambda - 1) = 0$, so $\lambda \in \{0, 1\}$.

Exercise 1.3 [Medium]

Let $u, v \in \mathbb{R}^n$ be orthogonal unit vectors (i.e., $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 0$). Define

$$A = 3uu^T + 5vv^T.$$

a) Prove that A is symmetric.
b) Find all eigenvalues and their corresponding eigenspaces.
c) What is $\operatorname{rank}(A)$?
d) Compute $\operatorname{tr}(A)$, $\det(A)$, and all singular values of A.

Solution

a) $A^T = 3(uu^T)^T + 5(vv^T)^T = 3uu^T + 5vv^T = A$. $\square$
b) The matrix A acts on three types of vectors:
Case 1: $x = u$. Then $Au = 3uu^Tu + 5vv^Tu = 3u \cdot 1 + 5v \cdot 0 = 3u$. So $\lambda_1 = 3$ with eigenvector u.
Case 2: $x = v$. Then $Av = 3uu^Tv + 5vv^Tv = 0 + 5v = 5v$. So $\lambda_2 = 5$ with eigenvector v.
Case 3: $x \in \operatorname{span}(u, v)^\perp$. Then $Ax = 3u(u^Tx) + 5v(v^Tx) = 0$. So $\lambda_3 = 0$ with eigenspace $\operatorname{span}(u, v)^\perp$ of dimension $n - 2$.
Eigenvalues: 5 (multiplicity 1), 3 (multiplicity 1), 0 (multiplicity $n - 2$).
c) $\operatorname{rank}(A) = 2$ (number of nonzero eigenvalues, since A is symmetric).
d) $\operatorname{tr}(A) = 5 + 3 + 0 \cdot (n - 2) = 8$.
$\det(A) = 5 \cdot 3 \cdot 0^{n-2} = 0$.
Since A is symmetric PSD (all eigenvalues ≥ 0), the singular values equal the eigenvalues: $\sigma_1 = 5$, $\sigma_2 = 3$, and $\sigma_k = 0$ for $k \geq 3$.

Exercise 1.4 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ and suppose A is diagonalizable with eigenvalues $\lambda_1, \ldots, \lambda_n$.

a) Prove that A^2 has eigenvalues $\lambda_1^2, \ldots, \lambda_n^2$.

b) More generally, show that for any polynomial $p(t) = \sum_{k=0}^m c_k t^k$, the matrix $p(A)$ has eigenvalues $p(\lambda_1), \ldots, p(\lambda_n)$.

c) Suppose A is invertible. Prove that A^{-1} has eigenvalues $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$.

Solution

a) Since A is diagonalizable, $A = PDP^{-1}$ where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then:

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$$

where $D^2 = \operatorname{diag}(\lambda_1^2, \ldots, \lambda_n^2)$. So A^2 is diagonalizable with eigenvalues λ_i^2.
b) By induction from part (a), $A^k = PD^kP^{-1}$ for any $k \geq 0$. Then:

$$p(A) = \sum_{k=0}^m c_k A^k = \sum_{k=0}^m c_k PD^kP^{-1} = P\left(\sum_{k=0}^m c_k D^k\right)P^{-1} = P \cdot \operatorname{diag}(p(\lambda_1), \ldots, p(\lambda_n)) \cdot P^{-1}$$

So $p(A)$ has eigenvalues $p(\lambda_i)$.
c) Since A is invertible, all $\lambda_i \neq 0$. From $A = PDP^{-1}$:

$$A^{-1} = PD^{-1}P^{-1} = P\operatorname{diag}(\lambda_1^{-1}, \ldots, \lambda_n^{-1})P^{-1}$$

So A^{-1} has eigenvalues λ_i^{-1}.

Exercise 1.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be a matrix such that $A^2 = A$ (i.e., A is idempotent).

a) Prove that A is diagonalizable. *Hint: show that* $\mathbb{R}^n = \ker(A) \oplus \operatorname{im}(A)$.

b) Prove that $\operatorname{tr}(A) = \operatorname{rank}(A)$.

c) Now suppose additionally that A is symmetric. Prove that A is an orthogonal projection, i.e., $A = A^T = A^2$, and that all eigenvalues are exactly 0 or 1.

Solution

a) We show $\mathbb{R}^n = \ker(A) \oplus \operatorname{im}(A)$.
Step 1 (Decomposition): For any $x \in \mathbb{R}^n$, write $x = (x - Ax) + Ax$. We have $Ax \in \operatorname{im}(A)$ and $A(x - Ax) = Ax - A^2x = Ax - Ax = 0$, so $(x - Ax) \in \ker(A)$. Hence $\mathbb{R}^n = \ker(A) + \operatorname{im}(A)$.
Step 2 (Direct sum): If $x \in \ker(A) \cap \operatorname{im}(A)$, then $Ax = 0$ and $x = Ay$ for some y. So $x = Ay$ and $0 = Ax = A^2y = Ay = x$.
Since $\mathbb{R}^n = \ker(A) \oplus \operatorname{im}(A)$, choosing a basis from each subspace gives a basis of eigenvectors: eigenvalue 0 on $\ker(A)$, eigenvalue 1 on $\operatorname{im}(A)$ (since $v \in \operatorname{im}(A)$ means $v = Aw$, so $Av = A^2w = Aw = v$). So A is diagonalizable.
b) Since A is diagonalizable with eigenvalues only 0 and 1:

$$\operatorname{tr}(A) = (\text{number of eigenvalues equal to } 1) = \dim(\operatorname{im}(A)) = \operatorname{rank}(A).$$

c) We already know from part (a) that $A^2 = A$ and the eigenvalues are 0 and 1. If A is also symmetric, then by the spectral theorem $A = Q\Lambda Q^T$ where Λ has entries in $\{0, 1\}$. Then $A^T = (Q\Lambda Q^T)^T = Q\Lambda Q^T = A$ (confirming symmetry), and $A^2 = Q\Lambda^2 Q^T = Q\Lambda Q^T = A$ since $\lambda_i^2 = \lambda_i$ for $\lambda_i \in \{0, 1\}$. This shows A is an orthogonal projection.

Symmetric Matrices and Positive (Semi-)Definiteness

Exercise 2.1 [Easy]

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

a) Verify that A is symmetric.

b) Find the eigenvalues of A. Is A positive definite, positive semi-definite, or neither?

c) Find an orthogonal matrix Q such that $A = Q\Lambda Q^T$.

Solution

a) $A^T = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = A$. ✓
b) $\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 5\lambda + 5 = 0$.
$\lambda = \frac{5 \pm \sqrt{5}}{2}$. Both eigenvalues are positive (since $\frac{5-\sqrt{5}}{2} \approx 1.38 > 0$), so A is **positive definite**.
c) For $\lambda_1 = \frac{5+\sqrt{5}}{2}$: solve $(A - \lambda_1 I)v = 0$ to get v_1, then normalize.
For $\lambda_2 = \frac{5-\sqrt{5}}{2}$: similarly find and normalize v_2.
Since A is symmetric, $v_1 \perp v_2$ automatically. Set $Q = (v_1 \mid v_2)$ and $\Lambda = \operatorname{diag}(\lambda_1, \lambda_2)$.

Exercise 2.2 [Easy]

True or False (with proof or counterexample):

a) If every entry of $A \in \mathbb{R}^{n\times n}$ is positive, then A is positive definite.

b) If A is positive semi-definite, then every diagonal entry $a_{ii} \geq 0$.
c) If A and B are both positive semi-definite, then $A + B$ is positive semi-definite.

Solution

a) False. Counterexample: $A = \begin{pmatrix} 1 & 10 \\ 10 & 1 \end{pmatrix}$. All entries are positive, but $\det(A) = 1 - 100 = -99 < 0$, so A has a negative eigenvalue and is not PSD.
b) True. If $A \succeq 0$, then for the standard basis vector e_i: $a_{ii} = e_i^T A e_i \geq 0$.
c) True. For any $x \in \mathbb{R}^n$: $x^T(A + B)x = x^T Ax + x^T Bx \geq 0 + 0 = 0$.

Exercise 2.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive definite, and let $B \in \mathbb{R}^{n\times m}$ with $\operatorname{rank}(B) = m$ (i.e., B has full column rank, with $m \leq n$).
a) Prove that $B^T AB \in \mathbb{R}^{m\times m}$ is symmetric.
b) Prove that $B^T AB$ is positive definite.
c) Conclude that $B^T AB$ is invertible.

Solution

a) $(B^T AB)^T = B^T A^T (B^T)^T = B^T AB$ since $A^T = A$. $\square$
b) For any $x \in \mathbb{R}^m$, $x \neq 0$, let $y = Bx \in \mathbb{R}^n$. Since B has full column rank, $\ker(B) = \{0\}$, so $y = Bx \neq 0$. Then:

$$x^T(B^T AB)x = (Bx)^T A(Bx) = y^T Ay > 0$$

where the last inequality uses that A is positive definite and $y \neq 0$. $\square$
c) Since $B^T AB$ is positive definite, all its eigenvalues are strictly positive. In particular, 0 is not an eigenvalue, so $B^T AB$ is invertible.

Exercise 2.4 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$.
a) Show that A^2 is positive semi-definite.
b) Show that if A is positive definite, then A^{-1} is also positive definite.
c) Show that if A is positive semi-definite, then $\operatorname{tr}(A) \geq 0$, and equality holds if and only if $A = 0$.

Solution

a) For any $x \in \mathbb{R}^n$: $x^T A^2 x = x^T A^T Ax = (Ax)^T(Ax) = \|Ax\|^2 \geq 0$. $\square$
b) If $A \succ 0$, all eigenvalues $\lambda_i > 0$. By the spectral theorem, $A = Q\Lambda Q^T$, so $A^{-1} = Q\Lambda^{-1}Q^T$ where $\Lambda^{-1} = \operatorname{diag}(\lambda_1^{-1}, \ldots, \lambda_n^{-1})$. Since each $\lambda_i^{-1} > 0$, A^{-1} is positive definite.
c) $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$. Since $A \succeq 0$, all $\lambda_i \geq 0$, so $\operatorname{tr}(A) = \sum \lambda_i \geq 0$. Equality holds iff all $\lambda_i = 0$, which for a symmetric matrix means $A = Q \cdot 0 \cdot Q^T = 0$.

Exercise 2.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive semi-definite. Define the map $\langle \cdot, \cdot \rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle x, y \rangle_A = x^T Ay.$$

a) Show that $\langle \cdot, \cdot \rangle_A$ satisfies symmetry, bilinearity, and non-negativity.

b) Under what additional condition on A is $\langle \cdot, \cdot \rangle_A$ an inner product? Prove your claim.
c) Now assume $A \succ 0$. Prove the generalized Cauchy-Schwarz inequality:

$$|x^T Ay|^2 \leq (x^T Ax)(y^T Ay)$$

for all $x, y \in \mathbb{R}^n$. *Hint: Since $A \succ 0$, there exists a unique $A^{1/2} \succ 0$ with $A = A^{1/2}A^{1/2}$.*

Solution

a) *Symmetry:* $\langle x, y \rangle_A = x^T Ay = (y^T A^T x)^T = (y^T Ax)^T = y^T Ax = \langle y, x \rangle_A$ (using $A = A^T$ and that the result is a scalar).
Bilinearity: $\langle \alpha x_1 + \beta x_2, y \rangle_A = (\alpha x_1 + \beta x_2)^T Ay = \alpha x_1^T Ay + \beta x_2^T Ay = \alpha \langle x_1, y \rangle_A + \beta \langle x_2, y \rangle_A$.
By symmetry, linearity in the second argument follows.
Non-negativity: $\langle x, x \rangle_A = x^T Ax \geq 0$ since $A \succeq 0$.
b) We need positive definiteness: $\langle x, x \rangle_A = 0 \implies x = 0$. This holds if and only if A is **positive definite** ($A \succ 0$).
Proof ($\Leftarrow$): If $A \succ 0$ and $x^T Ax = 0$, write $A = Q\Lambda Q^T$ with all $\lambda_i > 0$. Let $z = Q^T x$. Then $0 = z^T \Lambda z = \sum \lambda_i z_i^2$. Since $\lambda_i > 0$, all $z_i = 0$, so $x = Qz = 0$.
Proof ($\Rightarrow$): If A is only PSD with $\ker(A) \neq \{0\}$, pick $0 \neq x \in \ker(A)$. Then $\langle x, x \rangle_A = x^T Ax = 0$ but $x \neq 0$, so positive definiteness of the form fails.
c) Since $A \succ 0$, let $A^{1/2}$ be the unique positive definite square root. Then $x^T Ay = x^T A^{1/2} A^{1/2} y = \langle A^{1/2}x, A^{1/2}y \rangle$.
Applying the standard Cauchy-Schwarz inequality to $u = A^{1/2}x$ and $w = A^{1/2}y$:

$$|\langle u, w \rangle|^2 \leq \|u\|^2 \|w\|^2$$

$$|x^T Ay|^2 \leq (x^T A^{1/2} A^{1/2} x)(y^T A^{1/2} A^{1/2} y) = (x^T Ax)(y^T Ay). \qquad \square$$

Spectral Theorem and Variational Characterization

Exercise 3.1 [Easy]

Let $A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$.

a) Find the eigenvalues $\lambda_1 \leq \lambda_2$ of A.

b) Compute the Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$ for $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Verify that $\lambda_1 \leq R_A(x) \leq \lambda_2$.

c) What is $\max_{\|x\|=1} x^T Ax$ and which x achieves it?

Solution

a) $\det(A - \lambda I) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6)$.
So $\lambda_1 = 1$, $\lambda_2 = 6$.
b) $x^T Ax = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = 11$, and $x^T x = 2$.
$R_A(x) = \frac{11}{2} = 5.5$. Check: $1 \leq 5.5 \leq 6$. ✓
c) By the variational characterization, $\max_{\|x\|=1} x^T Ax = \lambda_{\max} = 6$, achieved by the unit eigenvector of $\lambda_2 = 6$.
For $\lambda_2 = 6$: $(A - 6I)v = 0$ gives $\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} v = 0$, so $v = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Exercise 3.2 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \le \lambda_2 \le \ldots \le \lambda_n$ and corresponding orthonormal eigenvectors $q_1, \ldots, q_n$.

a) Using the spectral decomposition $A = \sum_{i=1}^n \lambda_i q_i q_i^T$, show that

$$\max_{\|x\|=1} x^T A x = \lambda_n.$$

Hint: Write $x = \sum_{i=1}^n \alpha_i q_i$ *with* $\sum \alpha_i^2 = 1$.

b) Now consider the constrained problem:

$$\max_{\|x\|=1,\, x\perp q_n} x^T A x.$$

What is the maximum value? Prove your answer.

Solution

a) Write $x = \sum_{i=1}^n \alpha_i q_i$ with $\|x\|^2 = \sum \alpha_i^2 = 1$. Then, using orthonormality of the q_i:

$$x^T A x = \left(\sum_i \alpha_i q_i\right)^T \left(\sum_j \lambda_j \alpha_j q_j\right) = \sum_{i=1}^n \lambda_i \alpha_i^2.$$

Since $\sum \alpha_i^2 = 1$ and $\lambda_i \le \lambda_n$ for all i:

$$\sum_{i=1}^n \lambda_i \alpha_i^2 \le \lambda_n \sum_{i=1}^n \alpha_i^2 = \lambda_n.$$

Equality is achieved by $x = q_n$ (i.e., $\alpha_n = 1$, all others zero).

b) If $x \perp q_n$, then $\alpha_n = \langle x, q_n\rangle = 0$, so $x = \sum_{i=1}^{n-1} \alpha_i q_i$ with $\sum_{i=1}^{n-1} \alpha_i^2 = 1$. Then:

$$x^T A x = \sum_{i=1}^{n-1} \lambda_i \alpha_i^2 \le \lambda_{n-1} \sum_{i=1}^{n-1} \alpha_i^2 = \lambda_{n-1}.$$

Equality at $x = q_{n-1}$. So the constrained maximum is λ_{n-1}.

Exercise 3.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with spectral decomposition $A = Q\Lambda Q^T$. Define the matrix exponential $e^A = Q\,\mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})Q^T$.

a) Show that e^A is symmetric and positive definite.

b) Show that $\det(e^A) = e^{\mathrm{tr}(A)}$.

c) Show that $e^{A+B} = e^A e^B$ if A and B commute (i.e., $AB = BA$). *Hint: If $AB = BA$ and both are symmetric, they share an eigenbasis.*

Solution

a) *Symmetric:* $(e^A)^T = (Q\,\mathrm{diag}(e^{\lambda_i})Q^T)^T = Q\,\mathrm{diag}(e^{\lambda_i})Q^T = e^A$.
Positive definite: The eigenvalues of e^A are $e^{\lambda_1}, \ldots, e^{\lambda_n}$, and $e^{\lambda_i} > 0$ for all $\lambda_i \in \mathbb{R}$. Since all eigenvalues are strictly positive, $e^A \succ 0$.
b) $\det(e^A) = \prod_{i=1}^n e^{\lambda_i} = e^{\sum_{i=1}^n \lambda_i} = e^{\mathrm{tr}(A)}$.
(We used: determinant equals product of eigenvalues, and trace equals sum of eigenvalues.)
c) If A and B are symmetric and commute, they can be simultaneously diagonalized: there

exists orthogonal Q such that $A = Q\Lambda_A Q^T$ and $B = Q\Lambda_B Q^T$. Then $A+B = Q(\Lambda_A+\Lambda_B)Q^T$, so:

$$e^{A+B} = Q\,\mathrm{diag}(e^{\lambda_i^A+\lambda_i^B})Q^T = Q\,\mathrm{diag}(e^{\lambda_i^A}\cdot e^{\lambda_i^B})Q^T.$$

Meanwhile:

$$e^A e^B = Q\,\mathrm{diag}(e^{\lambda_i^A})Q^T \cdot Q\,\mathrm{diag}(e^{\lambda_i^B})Q^T = Q\,\mathrm{diag}(e^{\lambda_i^A}\cdot e^{\lambda_i^B})Q^T.$$

These are equal. □

Exercise 3.4 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite with eigenvalues $0 < \lambda_1 \le \ldots \le \lambda_n$.

a) Prove using the variational characterization that

$$\min_{\|x\|=1} \|Ax\|_2 = \lambda_1, \qquad \max_{\|x\|=1} \|Ax\|_2 = \lambda_n.$$

Hint: Consider $\|Ax\|_2^2 = x^T A^2 x$ and use the eigenvalues of A^2.

b) The condition number of A is defined as $\kappa(A) = \frac{\lambda_n}{\lambda_1}$. Show that $\kappa(A) = \|A\|_2\|A^{-1}\|_2$, where $\|A\|_2 = \max_{\|x\|=1}\|Ax\|_2$.

c) Solve the optimization problem

$$\max_{x\in\mathbb{R}^n,\ \|x\|_2=1} \|Ax\|_2 - \langle x, Ax\rangle.$$

Express the solution in terms of the eigenvalues of A.

Solution

a) $\|Ax\|_2^2 = x^T A^T A x = x^T A^2 x$ (since A is symmetric). The matrix A^2 is symmetric with eigenvalues $\lambda_1^2 \le \ldots \le \lambda_n^2$. By the Rayleigh quotient:

$$\max_{\|x\|=1} x^T A^2 x = \lambda_n^2 \implies \max_{\|x\|=1} \|Ax\|_2 = \lambda_n.$$

Similarly, $\min_{\|x\|=1}\|Ax\|_2 = \lambda_1$.

b) From part (a), $\|A\|_2 = \lambda_n$. The eigenvalues of A^{-1} are $\lambda_n^{-1} \le \ldots \le \lambda_1^{-1}$, so $\|A^{-1}\|_2 = \lambda_1^{-1}$. Thus:

$$\|A\|_2\|A^{-1}\|_2 = \frac{\lambda_n}{\lambda_1} = \kappa(A).$$

c) Write $x = \sum_{i=1}^n \alpha_i q_i$ with $\sum \alpha_i^2 = 1$. Then:

$$\|Ax\|_2 = \sqrt{\sum \lambda_i^2\alpha_i^2}, \qquad \langle x, Ax\rangle = \sum \lambda_i \alpha_i^2.$$

Define $g(\alpha) = \sqrt{\sum \lambda_i^2\alpha_i^2} - \sum \lambda_i\alpha_i^2$.
At any eigenvector $x = q_i$: $g = \lambda_i - \lambda_i = 0$.
For a mixture $x = \cos\theta\cdot q_1 + \sin\theta\cdot q_n$ ($\alpha_1 = \cos\theta$, $\alpha_n = \sin\theta$):

$$g = \sqrt{\lambda_1^2\cos^2\theta + \lambda_n^2\sin^2\theta} - (\lambda_1\cos^2\theta + \lambda_n\sin^2\theta).$$

By the inequality $\sqrt{a} \ge a/\sqrt{\max}$ for sums of positive terms, the maximum is $\frac{(\lambda_n-\lambda_1)^2}{4\lambda_n}$ achieved when $\sin^2\theta = \frac{\lambda_1}{2\lambda_n}$ (obtained by calculus). In general, the answer depends on the spectral gap: larger gap between λ_1 and λ_n means a larger maximum.

Singular Value Decomposition and Low-Rank Perturbations

Exercise 4.1 [Easy]

Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \\ 0 & 0 \end{pmatrix}$.

a) Compute $A^T A$ and AA^T.
b) Find the singular values of A.
c) Write down the full SVD $A = U\Sigma V^T$.

Solution

a) $A^T A = \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix}$, $\quad AA^T = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

b) Eigenvalues of $A^T A$: $\lambda_1 = 9$, $\lambda_2 = 4$. Singular values: $\sigma_1 = 3$, $\sigma_2 = 2$.
c) Right singular vectors (eigenvectors of $A^T A$): $v_1 = e_1$, $v_2 = e_2$ in $\mathbb{R}^2$.

Left singular vectors: $u_i = \frac{Av_i}{\sigma_i}$. So $u_1 = \frac{1}{3}\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = e_1$, $u_2 = \frac{1}{2}\begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} = -e_2$, and $u_3 = e_3$.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^T$$

Note: The sign of A's second column is negative, which is absorbed into U (since Σ must have non-negative entries).

Exercise 4.2 [Easy]

Let $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ be nonzero vectors. Consider the rank-one matrix $A = uv^T$.

a) What is $\text{rank}(A)$?
b) Find all singular values of A.
c) What are $\|A\|_2$ and $\|A\|_F$?

Solution

a) $\text{rank}(A) = 1$ since every column is a scalar multiple of u.
b) $A^T A = (uv^T)^T(uv^T) = vu^T uv^T = \|u\|^2 vv^T$. This is a rank-one matrix with one nonzero eigenvalue $\|u\|^2\|v\|^2$ (eigenvector v). So the only nonzero singular value is $\sigma_1 = \|u\| \cdot \|v\|$.
c) $\|A\|_2 = \sigma_{\max} = \|u\| \cdot \|v\|$.
$\|A\|_F = \sqrt{\sum \sigma_i^2} = \sigma_1 = \|u\| \cdot \|v\|$.
(For rank-one matrices, the operator norm and Frobenius norm coincide.)

Exercise 4.3 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$ (not necessarily non-negative).

a) Explain the relationship between the SVD of A and its spectral decomposition. How do the singular values relate to the eigenvalues?

b) Show that $\|A\|_F^2 = \sum_{i=1}^n \lambda_i^2$.

c) Let A_k denote the best rank-k approximation to A (in Frobenius norm). Express $\|A - A_k\|_F^2$ in terms of the eigenvalues of A.

Solution

a) For symmetric A, the spectral decomposition gives $A = Q\Lambda Q^T$ where $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. The singular values are the absolute values of the eigenvalues: $\sigma_i = |\lambda_i|$. The SVD can be obtained by reordering so that singular values are non-increasing and absorbing signs into U: if $\lambda_i < 0$, the corresponding column of U gets a sign flip relative to Q.
b) $\|A\|_F^2 = \text{tr}(A^T A) = \text{tr}(A^2)$ (since A is symmetric). The eigenvalues of A^2 are λ_i^2, so $\text{tr}(A^2) = \sum_{i=1}^n \lambda_i^2$.
c) By the Eckart-Young theorem, $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$ (using the top k singular values in decreasing order). Then:

$$\|A - A_k\|_F^2 = \sum_{i=k+1}^{n} \sigma_i^2 = \sum_{i=k+1}^{n} |\lambda_{\pi(i)}|^2$$

where π is the permutation that orders eigenvalues by decreasing absolute value: $|\lambda_{\pi(1)}| \geq \ldots \geq |\lambda_{\pi(n)}|$.

Exercise 4.4 [Medium]

Let $A \in \mathbb{R}^{m\times n}$ have rank r with singular values $\sigma_1 \geq \ldots \geq \sigma_r > 0$. Let $E \in \mathbb{R}^{m\times n}$ be a perturbation with $\text{rank}(E) = p$.

a) Using the low-rank perturbation bound, we know that

$$\sigma_{k+p}(A) \leq \sigma_k(A - E) \quad \text{for } k = 1, \ldots, r - p.$$

Explain in words what this means.

b) Suppose A has rank $r = 5$ with singular values $\sigma_1 = 10, \sigma_2 = 8, \sigma_3 = 5, \sigma_4 = 3, \sigma_5 = 1$ and we subtract a rank-2 matrix E. What can you conclude about the singular values of $A - E$?

c) Conversely, show that for the best rank-k approximation A_k, we have $\|A - A_k\|_2 = \sigma_{k+1}(A)$.

Solution

a) The bound says: subtracting a rank-p matrix can reduce each singular value by at most p positions in the ordering. The $(k + p)$-th singular value of the original matrix A serves as a lower bound on the k-th singular value of the perturbed matrix $A - E$. This is a key result for understanding how well low-rank approximations preserve singular values.
b) With $p = 2$, the bound gives:
$\sigma_3(A) = 5 \leq \sigma_1(A - E)$: the largest singular value of $A - E$ is at least 5.
$\sigma_4(A) = 3 \leq \sigma_2(A - E)$: the second singular value is at least 3.
$\sigma_5(A) = 1 \leq \sigma_3(A - E)$: the third singular value is at least 1.
Also, $\text{rank}(A - E) \geq r - p = 3$ since those singular values are bounded away from zero.
c) $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$, so $A - A_k = \sum_{i=k+1}^r \sigma_i u_i v_i^T$. This is already in SVD form (the u_i, v_i are orthonormal), with singular values $\sigma_{k+1} \geq \ldots \geq \sigma_r$. The operator norm equals the largest singular value:

$$\|A - A_k\|_2 = \sigma_{k+1}(A). \qquad \square$$

Exercise 4.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric PSD with eigenvalues $\lambda_1 \geq \ldots \geq \lambda_n \geq 0$ and orthonormal eigenvectors $q_1, \ldots, q_n$. Consider the rank-one update $B = A + \gamma q_1 q_1^T$ for some $\gamma > 0$.

a) Find all eigenvalues and eigenvectors of B.

b) How do the singular values of B relate to those of A?

c) Now consider a general rank-one update $C = A + \gamma ww^T$ where $w \in \mathbb{R}^n$ is a unit vector not aligned with any eigenvector of A. Without computing explicitly, explain why the eigenvalues of C interlace with those of A:

$$\mu_1 \geq \lambda_1 \geq \mu_2 \geq \lambda_2 \geq \ldots \geq \mu_n \geq \lambda_n$$

where μ_i are the eigenvalues of C in decreasing order.

Solution

a) Since $q_1, \ldots, q_n$ are eigenvectors of A and $q_1 q_1^T q_j = q_1 \delta_{1j}$:
$Bq_1 = Aq_1 + \gamma q_1 q_1^T q_1 = \lambda_1 q_1 + \gamma q_1 = (\lambda_1 + \gamma) q_1$.
For $j \geq 2$: $Bq_j = Aq_j + \gamma q_1 (q_1^T q_j) = \lambda_j q_j + 0 = \lambda_j q_j$.
Eigenvalues of B: $\lambda_1 + \gamma, \lambda_2, \ldots, \lambda_n$ with the same eigenvectors.
b) Since B is symmetric PSD (all eigenvalues ≥ 0), singular values equal eigenvalues: $\sigma_1(B) = \lambda_1 + \gamma$, $\sigma_i(B) = \lambda_i$ for $i \geq 2$. Only the top singular value increased by γ.
c) This is Cauchy's interlacing theorem for rank-one perturbations. The key argument uses the matrix determinant lemma. The characteristic polynomial of $C = A + \gamma ww^T$ satisfies:

$$\det(C - \mu I) = \det(A - \mu I) \left(1 + \gamma w^T (A - \mu I)^{-1} w\right).$$

The function $f(\mu) = 1 + \gamma \sum_{i=1}^n \frac{\alpha_i^2}{\lambda_i - \mu}$ (where $\alpha_i = q_i^T w$) is a decreasing function of μ between consecutive eigenvalues λ_i of A. It goes from $+\infty$ to $-\infty$ in each interval $(\lambda_{i+1}, \lambda_i)$. By the intermediate value theorem, there is exactly one root μ_i in each interval, producing the interlacing pattern.
Since $\gamma > 0$, all eigenvalues shift up or stay: $\mu_i \geq \lambda_i$ for all i.

Power Method

Exercise 5.1 [Easy]

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and starting vector $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

a) Perform 2 iterations of the power method: compute $x_1 = \frac{Ax_0}{\|Ax_0\|}$ and $x_2 = \frac{Ax_1}{\|Ax_1\|}$.

b) Estimate the dominant eigenvalue using the Rayleigh quotient $R_A(x_k)$ at each step.

c) What are the true eigenvalues? To which eigenvector does the power method converge?

Solution

a) $Ax_0 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\|Ax_0\| = \sqrt{5}$. So $x_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

$Ax_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, $\|Ax_1\| = \frac{\sqrt{41}}{\sqrt{5}}$. So $x_2 = \frac{1}{\sqrt{41}} \begin{pmatrix} 5 \\ 4 \end{pmatrix}$.

b) $R_A(x_0) = \frac{x_0^T A x_0}{x_0^T x_0} = 2$.

$R_A(x_1) = x_1^T A x_1 = \frac{1}{5}(2, 1) \begin{pmatrix} 5 \\ 4 \end{pmatrix} = \frac{14}{5} = 2.8$.

$R_A(x_2) = x_2^T A x_2 = \frac{1}{41}(5, 4) \begin{pmatrix} 14 \\ 13 \end{pmatrix} = \frac{122}{41} \approx 2.976$.

The estimates are converging toward 3.

c) Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$, with eigenvectors $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$ and $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$.

The power method converges to the dominant eigenvector $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$. We can see $x_2 = \frac{1}{\sqrt{41}}\begin{pmatrix}5\\4\end{pmatrix}$ is already approaching equal components.

Exercise 5.2 [Medium]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $|\lambda_1| > |\lambda_2| \geq \ldots \geq |\lambda_n|$ and orthonormal eigenvectors $q_1, \ldots, q_n$.

a) Write the starting vector $x_0 = \sum_{i=1}^n \alpha_i q_i$ with $\alpha_1 \neq 0$. Show that after k multiplications (without normalization):

$$A^k x_0 = \lambda_1^k \left(\alpha_1 q_1 + \sum_{i=2}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k q_i \right).$$

b) Explain why $\frac{A^k x_0}{\|A^k x_0\|} \to \pm q_1$ as $k \to \infty$.

c) What is the rate of convergence? What happens if $|\lambda_1| = |\lambda_2|$?

Solution

a) Since $Aq_i = \lambda_i q_i$, we have $A^k q_i = \lambda_i^k q_i$. Therefore:

$$A^k x_0 = \sum_{i=1}^n \alpha_i \lambda_i^k q_i = \lambda_1^k \sum_{i=1}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k q_i = \lambda_1^k \left(\alpha_1 q_1 + \sum_{i=2}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^k q_i \right).$$

We factored out λ_1^k so that we can see which terms vanish.

b) Since $|\lambda_1| > |\lambda_i|$ for $i \geq 2$, we have $\left|\frac{\lambda_i}{\lambda_1}\right| < 1$, so $\left(\frac{\lambda_i}{\lambda_1}\right)^k \to 0$ as $k \to \infty$. Thus:

$$A^k x_0 \approx \lambda_1^k \alpha_1 q_1 \quad \text{for large } k.$$

After normalization: $\frac{A^k x_0}{\|A^k x_0\|} \to \frac{\alpha_1}{|\alpha_1|} q_1 = \pm q_1$ (depending on the sign of α_1 and whether k is even/odd when $\lambda_1 < 0$).

c) The convergence rate is governed by $\left|\frac{\lambda_2}{\lambda_1}\right|^k$. The error after k steps is $O\left(\left|\frac{\lambda_2}{\lambda_1}\right|^k\right)$. This is **linear convergence** with rate $r = \left|\frac{\lambda_2}{\lambda_1}\right|$.

If $|\lambda_1| = |\lambda_2|$, the ratio $\left|\frac{\lambda_2}{\lambda_1}\right| = 1$, and the terms for q_2 do not decay. The method **fails to converge** (it may oscillate or converge to a subspace if there is a degenerate dominant eigenvalue).

Exercise 5.3 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 > \lambda_2 > \ldots > \lambda_n$. Suppose we already know λ_1 and q_1 (from the power method). We want to find λ_2 and q_2.

a) Define $B = A - \lambda_1 q_1 q_1^T$ (deflation). Show that B has eigenvalues $0, \lambda_2, \ldots, \lambda_n$ with the same eigenvectors as A.

b) Why can we not always directly apply the power method to B to find λ_2? What condition on λ_2 relative to the other eigenvalues is needed?

c) Propose a modification: use the *inverse power method* with shift μ close to λ_2, i.e., iterate $y_{k+1} = (B - \mu I)^{-1} y_k$ (normalized). Explain why this converges to the eigenvector for λ_2 and what determines the convergence rate.

Solution

a) For q_1: $Bq_1 = Aq_1 - \lambda_1 q_1 q_1^T q_1 = \lambda_1 q_1 - \lambda_1 q_1 = 0$. So q_1 is an eigenvector of B with eigenvalue 0.
For q_j with $j \geq 2$: $Bq_j = Aq_j - \lambda_1 q_1 (q_1^T q_j) = \lambda_j q_j - 0 = \lambda_j q_j$. So the remaining eigenvectors and eigenvalues are unchanged. $\square$
b) The power method on B converges to the eigenvector of the eigenvalue with *largest magnitude*. The eigenvalues of B are $0, \lambda_2, \ldots, \lambda_n$. If $|\lambda_n| > |\lambda_2|$ (e.g., if λ_n is very negative), the power method would converge to q_n, not q_2.
It works only if $|\lambda_2| > |\lambda_j|$ for all $j \geq 3$ (i.e., λ_2 has the largest magnitude among the remaining eigenvalues).
c) The matrix $(B - \mu I)^{-1}$ has eigenvalues $\frac{1}{\lambda_i(B) - \mu}$ for each eigenvalue $\lambda_i(B)$ of B. If μ is close to λ_2, then $\frac{1}{\lambda_2 - \mu}$ has the largest magnitude among all these eigenvalues, so the standard power method applied to $(B - \mu I)^{-1}$ converges to q_2.
Convergence rate: $\left|\frac{\lambda_2 - \mu}{\lambda_j - \mu}\right|$ for the nearest other eigenvalue $\lambda_j \neq \lambda_2$. The closer μ is to λ_2 (and the farther from other eigenvalues), the faster the convergence. In practice, each step requires solving a linear system $(B - \mu I) y_{k+1} = y_k$.

Pseudo-Inverse, Cholesky, Generalized Eigenproblems, Von Neumann

Exercise 6.1 [Easy]

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ (full column rank).

a) Compute the pseudo-inverse $A^\dagger$ using the formula $A^\dagger = (A^T A)^{-1} A^T$.
b) Verify that $A^\dagger A = I_2$.
c) What is $AA^\dagger$? Is it an orthogonal projection? Onto what subspace?

Solution

a) $A^T A = I_2$. So $A^\dagger = (I_2)^{-1} A^T = A^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.
b) $A^\dagger A = A^T A = I_2$. ✓
c) $AA^\dagger = AA^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.
This is the orthogonal projection onto $\mathrm{im}(A) = \mathrm{span}(e_1, e_2) \subset \mathbb{R}^3$ (the $x_1 x_2$-plane). We can verify: $(AA^\dagger)^2 = AA^\dagger$ and $(AA^\dagger)^T = AA^\dagger$.

Exercise 6.2 [Medium]

Let $A = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$.

a) Verify that A is positive definite.

b) Find the Cholesky factorization $A = LL^T$ where L is lower triangular with positive diagonal entries.

c) Use the Cholesky factorization to solve $Ax = b$ where $b = \begin{pmatrix} 8 \\ 11 \end{pmatrix}$. (First solve $Ly = b$, then $L^T x = y$.)

Solution

a) Eigenvalues: $\det(A - \lambda I) = (4-\lambda)(5-\lambda) - 4 = \lambda^2 - 9\lambda + 16 = 0$. So $\lambda = \frac{9 \pm \sqrt{81-64}}{2} = \frac{9 \pm \sqrt{17}}{2}$. Both are positive (since $\sqrt{17} < 9$), so $A \succ 0$.

b) $L = \begin{pmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{pmatrix}$. From $LL^T = A$:

$\ell_{11}^2 = 4 \implies \ell_{11} = 2$.

$\ell_{21}\ell_{11} = 2 \implies \ell_{21} = 1$.

$\ell_{21}^2 + \ell_{22}^2 = 5 \implies 1 + \ell_{22}^2 = 5 \implies \ell_{22} = 2$.

So $L = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$.

c) Solve $Ly = b$: $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} y = \begin{pmatrix} 8 \\ 11 \end{pmatrix}$. $y_1 = 4$, $y_2 = \frac{11-4}{2} = \frac{7}{2}$.

Solve $L^T x = y$: $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} x = \begin{pmatrix} 4 \\ 7/2 \end{pmatrix}$. $x_2 = \frac{7}{4}$, $x_1 = \frac{4 - 7/4}{2} = \frac{9}{8}$.

So $x = \begin{pmatrix} 9/8 \\ 7/4 \end{pmatrix}$.

Exercise 6.3 [Medium]

Let $A, B \in \mathbb{R}^{n \times n}$ with B symmetric positive definite. Consider the **generalized eigenvalue problem** $Ax = \lambda Bx$.

a) Show that this can be reduced to a standard eigenvalue problem by substituting $z = B^{1/2}x$. What is the resulting standard eigenproblem?

b) Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. Find the generalized eigenvalues.

c) Show that the generalized Rayleigh quotient $R_{A,B}(x) = \frac{x^T A x}{x^T B x}$ achieves its maximum at the largest generalized eigenvalue.

Solution

a) From $Ax = \lambda Bx$, multiply both sides on the left by $B^{-1/2}$:

$$B^{-1/2} A x = \lambda B^{1/2} x.$$

Substitute $x = B^{-1/2} z$ (equivalently $z = B^{1/2} x$):

$$B^{-1/2} A B^{-1/2} z = \lambda z.$$

This is a standard eigenvalue problem for the symmetric matrix $\tilde{A} = B^{-1/2} A B^{-1/2}$.

b) $B = 2I$, so $B^{-1/2} = \frac{1}{\sqrt{2}} I$. Then $\tilde{A} = \frac{1}{2} A = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}$.

Eigenvalues of $\tilde{A}$: $\frac{3}{2} \pm \frac{1}{2}$, giving $\lambda_1 = 1$ and $\lambda_2 = 2$.

(Equivalently, solve $\det(A - \lambda B) = 0$: $(3 - 2\lambda)^2 - 1 = 0 \implies \lambda = 1$ or $\lambda = 2$.)

c) Using the substitution $z = B^{1/2}x$:

$$R_{A,B}(x) = \frac{x^T Ax}{x^T Bx} = \frac{z^T B^{-1/2} A B^{-1/2} z}{z^T z} = R_{\tilde{A}}(z).$$

By the standard variational characterization, $\max_z R_{\tilde{A}}(z) = \lambda_{\max}(\tilde{A})$, which equals the largest generalized eigenvalue.

Exercise 6.4 [Medium]

(Von Neumann Trace Inequality) Let $A, B \in \mathbb{R}^{n\times n}$ with singular values $\sigma_1(A) \geq \ldots \geq \sigma_n(A)$ and $\sigma_1(B) \geq \ldots \geq \sigma_n(B)$ respectively. The von Neumann trace inequality states:

$$|\operatorname{tr}(A^T B)| \leq \sum_{i=1}^{n} \sigma_i(A)\sigma_i(B).$$

a) Verify this inequality for $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$.

b) Show that equality holds when A and B share the same left and right singular vectors, i.e., $A = U\Sigma_A V^T$ and $B = U\Sigma_B V^T$.

c) Use the von Neumann trace inequality to show that for any $A \in \mathbb{R}^{n\times n}$: $|\operatorname{tr}(A)| \leq \sum_{i=1}^{n} \sigma_i(A)$.

Solution

a) $\operatorname{tr}(A^T B) = \operatorname{tr}\left(\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}\right) = \operatorname{tr}\begin{pmatrix} 0 & 6 \\ 2 & 0 \end{pmatrix} = 0.$
So $|\operatorname{tr}(A^T B)| = 0$.
Singular values: $\sigma(A) = (3, 1)$, $\sigma(B) = (2, 2)$. $\sum \sigma_i(A)\sigma_i(B) = 3 \cdot 2 + 1 \cdot 2 = 8$.
$0 \leq 8$. ✓
b) If $A = U\Sigma_A V^T$ and $B = U\Sigma_B V^T$, then $A^T B = V\Sigma_A U^T U\Sigma_B V^T = V\Sigma_A\Sigma_B V^T$. So:

$$\operatorname{tr}(A^T B) = \operatorname{tr}(V\Sigma_A\Sigma_B V^T) = \operatorname{tr}(\Sigma_A\Sigma_B) = \sum_i \sigma_i(A)\sigma_i(B).$$

Thus equality holds.
c) Take $B = I_n$. Then $\sigma_i(I) = 1$ for all i. The inequality gives:

$$|\operatorname{tr}(A^T \cdot I)| = |\operatorname{tr}(A^T)| = |\operatorname{tr}(A)| \leq \sum_{i=1}^{n} \sigma_i(A) \cdot 1 = \sum_{i=1}^{n} \sigma_i(A). \qquad \square$$

Mixed Challenging Exercises

Exercise 7.1 [Hard]

Let $u, v \in \mathbb{R}^n$ with $\|u\| = 2$, $\|v\| = 1$, and $\langle u, v\rangle = 1$. Define

$$A = uu^T - vv^T.$$

a) Prove that A is symmetric. What is $\operatorname{rank}(A)$?

b) The matrix A acts nontrivially only on $\operatorname{span}(u, v)$. Find the 2×2 matrix representing A restricted to $\operatorname{span}(u, v)$ with respect to the basis $\{u, v\}$, and find all eigenvalues of A.

c) Compute $\mathrm{tr}(A)$, $\det(A)$, and all singular values of A.
d) Solve $\max_{x\in\mathbb{R}^n,\ \|x\|_2=1} \|Ax\|_2$.

Solution

a) $A^T = (uu^T)^T - (vv^T)^T = uu^T - vv^T = A$. ✓
A maps everything in $\mathrm{span}(u,v)^\perp$ to zero. Since u and v are not parallel ($|\langle u,v\rangle| = 1 < \|u\|\|v\| = 2$ by strict Cauchy-Schwarz), $\dim(\mathrm{span}(u,v)) = 2$, so $\mathrm{rank}(A) \le 2$. We determine the exact rank from the eigenvalues.
b) Compute A applied to the basis vectors:
$Au = uu^Tu - vv^Tu = \|u\|^2u - \langle v,u\rangle v = 4u - v$.
$Av = uu^Tv - vv^Tv = \langle u,v\rangle u - \|v\|^2 v = u - v$.
Representation in basis $\{u,v\}$ (column j = coordinates of A applied to j-th basis vector):

$$M = \begin{pmatrix} 4 & 1 \\ -1 & -1 \end{pmatrix}.$$

Note: this basis is *not* orthonormal, but the eigenvalues of M are the nonzero eigenvalues of A.
$\det(M - \lambda I) = (4-\lambda)(-1-\lambda) + 1 = \lambda^2 - 3\lambda - 3 = 0$.
$\lambda = \frac{3\pm\sqrt{21}}{2}$.
All eigenvalues of A: $\frac{3+\sqrt{21}}{2} \approx 3.79$, $\frac{3-\sqrt{21}}{2} \approx -0.79$, and 0 with multiplicity $n-2$.
$\mathrm{rank}(A) = 2$ (two nonzero eigenvalues).
c) $\mathrm{tr}(A) = \frac{3+\sqrt{21}}{2} + \frac{3-\sqrt{21}}{2} + 0 = 3$.
Alternatively: $\mathrm{tr}(A) = \mathrm{tr}(uu^T) - \mathrm{tr}(vv^T) = \|u\|^2 - \|v\|^2 = 4 - 1 = 3$. ✓
$\det(A) = 0$ (since eigenvalue 0 appears with multiplicity ≥ 1).
Singular values: $\sigma_i = |\lambda_i|$ for symmetric A. So $\sigma_1 = \frac{3+\sqrt{21}}{2}$, $\sigma_2 = \frac{\sqrt{21}-3}{2}$, and $\sigma_k = 0$ for $k \ge 3$.
d) $\max_{\|x\|=1} \|Ax\|_2 = \sigma_{\max}(A) = \frac{3+\sqrt{21}}{2}$, achieved at the unit eigenvector corresponding to eigenvalue $\frac{3+\sqrt{21}}{2}$.

Exercise 7.2 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite.

a) Show that A has a unique Cholesky decomposition $A = LL^T$ where L is lower triangular with positive diagonal entries. *Hint: Use induction on n with the block structure* $A = \begin{pmatrix} a_{11} & w^T \\ w & A_{22} \end{pmatrix}$.

b) Let A have eigenvalues $\lambda_1 \ge \ldots \ge \lambda_n > 0$. Express the condition number $\kappa(A)$ in terms of the singular values of the Cholesky factor L.

c) Show that the pseudo-inverse of A equals A^{-1} when A is invertible, by verifying all four Moore-Penrose conditions.

Solution

a) *Base case:* $n = 1$: $A = (a_{11})$ with $a_{11} > 0$, so $L = (\sqrt{a_{11}})$ is unique.
Inductive step: Write $A = \begin{pmatrix} a_{11} & w^T \\ w & A_{22} \end{pmatrix}$ with $a_{11} > 0$ (since $e_1^T A e_1 = a_{11} > 0$).
Set $\ell_{11} = \sqrt{a_{11}}$ (unique positive choice) and $l = w/\ell_{11}$. Then the Schur complement is $S = A_{22} - ll^T = A_{22} - ww^T/a_{11}$.

The Schur complement S is PD (standard result: for PD block matrices, the Schur complement is PD). By induction, $S = \tilde{L}\tilde{L}^T$ uniquely. Then $L = \begin{pmatrix} \ell_{11} & 0 \\ l & \tilde{L} \end{pmatrix}$ and $LL^T = A$. Each step is uniquely determined.
b) Since $A = LL^T$, the eigenvalues of A satisfy $\lambda_i(A) = \sigma_i(L)^2$ (this follows from $\sigma_i(L)^2 = \lambda_i(L^TL) = \lambda_i(A)$ since A is SPD and L^TL has the same eigenvalues as $LL^T = A$). So:

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} = \frac{\sigma_{\max}(L)^2}{\sigma_{\min}(L)^2} = \kappa(L)^2.$$

c) When A is invertible, set $A^\dagger = A^{-1}$. Verify the four Moore-Penrose conditions:
1. $AA^\dagger A = A \cdot A^{-1} \cdot A = A$. ✓
2. $A^\dagger AA^\dagger = A^{-1}AA^{-1} = A^{-1}$. ✓
3. $(AA^\dagger)^T = I^T = I = AA^{-1}$. ✓
4. $(A^\dagger A)^T = I^T = I = A^{-1}A$. ✓

All four conditions satisfied, so $A^\dagger = A^{-1}$.

Exercise 7.3 [Hard]

Let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $\operatorname{rank}(A) = n$ (full column rank). Consider the least squares problem $\min_x \|Ax - b\|_2^2$.

a) Show that the solution is $x^* = A^\dagger b = (A^TA)^{-1}A^Tb$.

b) Show that $P = AA^\dagger = A(A^TA)^{-1}A^T$ is an orthogonal projector onto $\operatorname{im}(A)$. Verify $P^2 = P$ and $P^T = P$.

c) Now suppose $A = U\Sigma V^T$ is the thin SVD (with $U \in \mathbb{R}^{m\times n}$, $\Sigma \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{n\times n}$). Express x^* and P in terms of U, Σ, V.

d) Show that the residual $r = b - Ax^*$ satisfies $r \perp \operatorname{im}(A)$.

Solution

a) $f(x) = \|Ax - b\|^2 = x^TA^TAx - 2b^TAx + b^Tb$. Setting $\nabla_x f = 2A^TAx - 2A^Tb = 0$ gives the normal equations $A^TAx = A^Tb$. Since $\operatorname{rank}(A) = n$, the matrix A^TA is invertible, so $x^* = (A^TA)^{-1}A^Tb = A^\dagger b$.
b) Idempotent: $P^2 = A(A^TA)^{-1}A^T \cdot A(A^TA)^{-1}A^T = A(A^TA)^{-1}(A^TA)(A^TA)^{-1}A^T = A(A^TA)^{-1}A^T = P$.
Symmetric: $P^T = (A(A^TA)^{-1}A^T)^T = A((A^TA)^{-1})^TA^T = A(A^TA)^{-1}A^T = P$.
(We used $(A^TA)^T = A^TA$ implies $((A^TA)^{-1})^T = (A^TA)^{-1}$.)
For any $y = Ac \in \operatorname{im}(A)$: $Py = A(A^TA)^{-1}A^TAc = Ac = y$. So P fixes $\operatorname{im}(A)$, confirming it projects onto $\operatorname{im}(A)$.
c) $A^\dagger = (A^TA)^{-1}A^T = (V\Sigma^2V^T)^{-1}V\Sigma U^T = V\Sigma^{-2}V^TV\Sigma U^T = V\Sigma^{-1}U^T$.
So $x^* = V\Sigma^{-1}U^Tb$.
$P = AA^\dagger = U\Sigma V^T \cdot V\Sigma^{-1}U^T = UU^T$.
d) $r = b - Ax^* = b - Pb = (I - P)b$. For any $w = Ac \in \operatorname{im}(A)$:

$$\langle r, w\rangle = ((I - P)b)^TAc = b^T(I - P)^TAc = b^T(I - P)Ac.$$

Since $(I - P)A = A - PA = A - A = 0$, we get $\langle r, w\rangle = 0$. So $r \perp \operatorname{im}(A)$. □

Exercise 7.4 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigenvalues $\lambda_1 \leq \ldots \leq \lambda_n$ and let $E = \gamma ww^T$ where $w \in \mathbb{R}^n$ is a unit vector and $\gamma > 0$.

a) What is $\text{rank}(E)$? What is $\|E\|_2$?

b) Let $\mu_1 \leq \ldots \leq \mu_n$ be the eigenvalues of $A+E$. Using the interlacing property for rank-one perturbations, give the tightest bounds you can on each μ_i.

c) Now suppose A has a spectral gap: $\lambda_n - \lambda_{n-1} = \Delta > 0$, and $\gamma < \Delta$. Show that μ_n remains the unique largest eigenvalue of $A+E$, and give a bound on $|\mu_n - \lambda_n|$.

d) Explain why this result is relevant for the convergence of the power method applied to $A+E$ versus A.

Solution

a) $\text{rank}(E) = 1$ (rank-one outer product). $\|E\|_2 = \gamma\|w\|^2 = \gamma$ (the only nonzero eigenvalue of E is γ).

b) By Cauchy interlacing for rank-one PSD perturbations:

$$\lambda_i \leq \mu_i \leq \lambda_{i+1} \quad \text{for } i = 1, \ldots, n-1$$

and $\lambda_n \leq \mu_n \leq \lambda_n + \gamma$ (from $\text{tr}(A+E) = \text{tr}(A) + \gamma$).
So each eigenvalue can increase by at most γ total, and the new eigenvalues interlace with the old ones.

c) From part (b): $\mu_{n-1} \leq \lambda_n$ and $\mu_n \geq \lambda_n$. We need $\mu_n > \mu_{n-1}$.
Since $\mu_{n-1} \leq \lambda_n$ and $\mu_n \leq \lambda_n + \gamma$, and also $\mu_{n-1} \geq \lambda_{n-1} = \lambda_n - \Delta$:
The gap satisfies $\mu_n - \mu_{n-1} \geq \lambda_n - \lambda_n = 0$, but generically (for w not orthogonal to q_n), $\mu_n > \lambda_n$, so the gap is strictly positive.
Bound: $|\mu_n - \lambda_n| \leq \|E\|_2 = \gamma$. Since $\gamma < \Delta$, we have $\mu_n - \mu_{n-1} \geq \Delta - \gamma > 0$, so μ_n remains the unique largest eigenvalue.

d) The power method converges at rate $\left|\frac{\mu_{n-1}}{\mu_n}\right|^k$ for $A+E$. Since the spectral gap is preserved (at least $\Delta - \gamma > 0$), the power method still converges, with a convergence rate close to $\left|\frac{\lambda_{n-1}}{\lambda_n}\right|^k$. This shows the power method is **robust to small perturbations**: if $\gamma \ll \Delta$, the convergence behavior barely changes.